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Systems and methods for client-controlled adaptive streaming of three-dimensional content

Abstract

A three-dimensional (3D) streaming system provides client-controlled adaptive streaming of 3D content in which the client may receive the 3D primitives of 3D content at different fidelities that the client selects. The system includes generating a 3D model at different fidelities, partitioning the 3D model into different volumetric spatial units, generating a manifest with an index for each volumetric spatial unit and an identifier for each fidelity, and presenting the 3D primitives in a first set of volumetric spatial units at a first fidelity and the 3D primitives in a second set of volumetric spatial units at a second fidelity in response to a request that includes indices of the first set of volumetric spatial units with a first identifier for the first fidelity and indices of the second set of volumetric spatial units with a second identifier for the second fidelity.

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Background/Summary

CLAIM OF BENEFIT TO RELATED APPLICATIONS (1) This application is a continuation of U.S. nonprovisional application Ser. No. 18/950,507 with the title “Systems and Methods for Client-Controlled Adaptive Streaming of Three-Dimensional Content”, filed Nov. 18, 2024, now issued as U.S. Pat. No. 12,262,099. The contents of application Ser. No. 18/950,507 are hereby incorporated by reference.

BACKGROUND

(1) Adaptive streaming protocols and techniques exists for the transmission of two-dimensional (2D) content or three-dimensional (3D) content that is converted to 2D content before it is streamed to a receiving client. The adaptive streaming protocols and techniques allow for the 2D content to be sent at different resolutions depending on the network conditions and/or available resources of the receiving client. Specifically, adaptive streaming of 2D content allows for all pixels of a first 2D frame or 2D image to be streamed at a first resolution and all pixels of a second 2D frame or 2D image to be stream at a different second resolution.

(2) Virtual reality, mixed reality, augmented reality, and other enhanced reality or spatial computing experiences as well as improvements to network capacity and client resources have created a need for native 3D content streaming. In other words, there is a need to stream the original 3D primitives

that form the 3D shapes and structures in 3D mesh models, point clouds, Gaussian splat 3D representations, and/or other 3D formats without first converting these 3D primitives or 3D formats to a 2D format or 2D image. The adaptive streaming protocols and techniques for 2D content do not apply and cannot be used to optimize the network delivery of the 3D primitive. Moreover, the 3D primitives of a 3D model are typically encoded with exponentially more data than a 2D image. As such, there is need for adaptive streaming protocols and techniques that apply and optimize the streaming of native 3D primitives or content over a data network.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a client-controlled adaptive streaming of three-dimensional (3D) content in accordance with some embodiments presented herein.
- (2) FIG. 2 illustrates an example of the volumetric spatial units that are generated for the client-controlled adaptive streaming in accordance with some embodiments presented herein.
- (3) FIG. 3 illustrates an example of generating 3D content at different fidelities in accordance with some embodiments presented herein.
- (4) FIG. 4 presents a process for the client-controlled adaptive streaming of 3D content in accordance with some embodiments presented herein.
- (5) FIG. 5 illustrates an example of the client-controlled adaptive streaming of new 3D primitives for a current field-of-view to combine with previously requested 3D primitives from a last field-of-view that remain in the current field-of-view in accordance with some embodiments presented herein.
- (6) FIG. 6 illustrates an example of the client-controlled adaptive streaming based on predicted user focus in accordance with some embodiments.
- (7) FIG. 7 illustrates an example of the client-controlled adaptive streaming with temporal values in accordance with some embodiments presented herein.
- (8) FIG. 8 illustrates an example of the client-controlled dynamic streaming with flagged volumetric spatial units in accordance with some embodiments presented herein.
- (9) FIG. 9 illustrates example components of one or more devices, according to one or more embodiments described herein.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

- (10) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.
- (11) Provided are systems and methods for client-controlled adaptive streaming of three-dimensional (3D) content. The client-controlled adaptive streaming includes streaming the 3D primitives of different 3D formats for a single scene or frame of static, dynamic, or animated 3D content at different fidelities that the client selects. Accordingly, the client-controlled adaptive streaming applies to meshes or polygons of a 3D mesh model, points of a point cloud, Gaussian splats of a 3D Gaussian splat representation, and/or other 3D primitives used to construct different 3D models.
- (12) The client-controlled adaptive streaming includes a server-side partitioning of the 3D space in which the 3D content is defined. The partitioning subdivides the 3D space into volumetric spatial units. A volumetric spatial unit is a non-overlapping subdivided region of the 3D space in which a subset of 3D primitives of the 3D content may be defined or positioned. Each volumetric spatial unit is assigned an index.
- (13) The client-controlled adaptive streaming includes generating a manifest that associates the index of each volumetric spatial unit to the different fidelities at which the 3D primitives in that volumetric spatial unit may be accessed. For instance, a lowest fidelity representation for a

volumetric spatial unit includes a base or minimum set of 3D primitives for representing the part of the 3D content in the corresponding region of 3D space represented by the volumetric spatial unit at a low level-of-detail or low visual quality. Increasing fidelities increase the number of 3D primitives, shrink the size of the 3D primitives for added detail, and/or increase density of the 3D primitives for representing the part of the 3D content in the corresponding region of 3D space represented by the volumetric spatial unit with more detail and/or better visual quality. In some embodiments, the 3D primitives of the 3D content may be associated with one of the different fidelities.

(14) The client receives the manifest in response to submitting a request for the 3D content to the server from which the 3D content is accessible. The client may request different volumetric spatial units that are listed in the manifest at different fidelities depending on the available device and network resources as well as which part of the 3D content receives the user focus. For instance, the client may request a first set of volumetric spatial units at the highest fidelity in response to the first set of volumetric spatial units corresponding to 3D content regions in the foreground and at the center of a current field-of-view or 3D content regions where the user is looking. The client may request volumetric spatial units that are more and more distant from the first set of volumetric spatial units at progressively lower fidelities to preserve network bandwidth and reduce the rendering load on the client while preserving visual quality on the user focused regions.

(15) FIG. 1 illustrates an example of the client-controlled adaptive streaming of 3D content in accordance with some embodiments presented herein. 3D streaming system **100** includes one or more devices or machines for hosting and/or streaming various 3D content across a data network to requesting clients. Accordingly, 3D streaming system **100** may include a 3D game streaming service, a 3D video streaming service, a 3D spatial computing experience service (e.g., virtual reality, mixed reality, augmented reality, or other enhanced reality service provider), and/or other distributor of 3D content.

(16) Client **101** submits (at **102**) a request for 3D content to 3D streaming system **100**. The request may include a HyperText Transfer Protocol (HTTP) GET message or another request specified using a different network protocol or request type. The request may be directed to a Uniform Resource Locator (URL) that includes the path and name for the requested 3D content.

(17) 3D streaming system **100** replies to the request by forwarding (at **104**) the manifest that is generated for the 3D content to client **101**. The manifest includes the indices for the different volumetric spatial units that partition the 3D space of the 3D content and the different fidelities at which each volumetric spatial unit may be accessed, requested, and/or rendered.

(18) Client **101** selectively requests (at **106**) different volumetric spatial units at different fidelities based on available network and/or rendering resources and a prioritization of the field-of-view from which the 3D content is presented. The initial field-of-view may be defined from a default placement and orientation of a virtual camera in the 3D space of the 3D content or from a last tracked placement and orientation of the virtual camera. In other words, whenever the 3D content is first requested by any client device, the 3D content may be presented from the same initial field-of-view and the user may then freely move within the 3D space (e.g., change the position and orientation of the virtual camera) to change the field-of-view and/or presentation of the 3D content.

(19) In some embodiments, selectively requesting (at **106**) the different volumetric spatial units may include requesting a higher fidelity for the volumetric spatial units at the center and/or foreground of the current field-of-view and requesting volumetric spatial units that are further removed from the center and the foreground at progressively lower and lower fidelities. In some embodiments, selectively requesting (at **106**) the different volumetric spatial units may include requesting a higher fidelity for the volumetric spatial units that receive user focus and requesting volumetric spatial units that are further removed from the user focus at progressively lower and lower fidelities. In some embodiments, selectively requesting (at **106**) the different volumetric spatial units may include requesting all volumetric spatial units within the initial field-of-view at

the lowest fidelity, monitoring resource availability, and then increasing fidelity for the volumetric spatial units at the center, in the foreground, and/or that receive the user focus if there are excess resources.

(20) 3D streaming system **100** streams (at **108**) the 3D primitives that are encoded in the requested (at **106**) volumetric spatial units at the specified fidelity. Accordingly, the 3D primitives for volumetric spatial units requested at a higher fidelity may be streamed at a higher level-of-detail, with a greater density of 3D primitives, and/or with a greater number of smaller sized 3D primitive than volumetric spatial units requested at a lower fidelity.

(21) Client **101** may track changes to the current field-of-view. The changes to the field-of-view may be in response to user input. In some embodiments, the user input may include a controller input for changing the field-of-view. The controller input may change the position of a virtual camera or a playable character in the 3D space. In some embodiments, the user input may include input that tracks the user focus. For instance, eye tracking sensors may track the user gaze and may change the field-of-view according to the tracked user gaze. Motion sensors may also track the user focus. For instance, sensors that track movement of a headset, the user body, or a device carried by the user may be used to change the field-of-view.

(22) Client **101** may selectively request different volumetric spatial units at different fidelities in response to the tracked changes to the field-of-view. For instance, the volumetric spatial units that move from the background to the foreground, from the field-of-view periphery to the center, or into the user focus may be requested at increased fidelity and the volumetric spatial units that move from the foreground to the background, from the center to the periphery, or out of the user focus may be requested at decreased fidelity or may be retained at the current fidelity to avoid redundant streaming of 3D primitives.

(23) FIG. 2 illustrates an example of the volumetric spatial units that are generated for the client-controlled adaptive streaming in accordance with some embodiments presented herein. 3D streaming system **100** receives (at **202**) 3D content. The 3D content may be defined with meshes or polygons of a 3D mesh model, points of a point cloud, Gaussian splats for a Gaussian splat representation, and/or 3D primitives of another 3D format.

(24) 3D streaming system **100** divides (at **204**) the 3D space in which the 3D content is defined into smaller volumetric spatial units. In some embodiments, each volumetric spatial unit may be the same size and/or span a similar volume of the 3D space in a non-overlapping manner. In some other embodiments, the volumetric spatial units may be defined to span a non-uniformly sized region of the 3D space so that an equal number of 3D primitives is contained within each of the volumetric spatial units.

(25) 3D streaming system **100** assigns (at **206**) a unique index to each volumetric spatial unit. The index for a volumetric spatial unit may be a numerical value or coordinate that maps to the region or volume of 3D space spanned by that volumetric spatial unit.

(26) 3D streaming system **100** enters (at **208**) the indices into the 3D content manifest. In some embodiments, the indices are arranged or positioned in the content manifest to directly map to the regions or volume of 3D space represented by the volumetric spatial unit identified by an index. For instance, the manifest may include the dimensions or size of each volumetric spatial unit and the indices are arranged to map volumetric spatial units of the defined size in the 3D space of the 3D content. Since volumetric spatial units may be defined at different depths, the manifest may arrange the indices as a multi-dimensional matrix with each dimension of the matrix corresponding to volumetric spatial units at a different depth in the 3D space. In some embodiments, each index may directly map to a volumetric spatial unit by defining the index with 8 coordinates for the corners of the volumetric spatial unit. In some other embodiments, the manifest includes a table that maps each index to the coordinates the volumetric spatial unit identified by that index.

(27) 3D streaming system **100** may encode the 3D content at different fidelities and may generate the manifest by associating the 3D primitives for the different fidelities at which the 3D content is

encoded to the indices of the volumetric spatial units in which the 3D primitives are located or positioned. FIG. 3 illustrates an example of generating 3D content at different fidelities in accordance with some embodiments presented herein.

(28) 3D streaming system **100** may receive (at **302**) input data for generating a 3D model. The input data may include 2D images that capture an object or scene from different perspectives, angles, and/or positions. The input data may include 3D scan data that is combined with visual data from a 2D image. For instance, a Light Detection and Ranging (LiDAR) or other 3D scanner may be used to obtain 3D positional data for different scanned points from an object or scene and the 2D image may capture the colors at the scanned points.

(29) 3D streaming system **100** may generate (at **304**) the 3D content at different fidelities from the input data. In some embodiments, 3D streaming system **100** uses a radiance field or neural network to generate Gaussian splats, points, or other 3D primitives based on the input data. In some such embodiments, 3D streaming system **100** generates the 3D model at the different fidelities in multiple iterative passes with each iterative pass adding more 3D primitives to increase the density and detail at which the 3D model is represented and with the 3D primitives generated in each pass corresponding to a different fidelity representation of the 3D model. For instance, 3D streaming system **100** may configure a Gaussian splatting engine with a first amount of acceptable loss for the loss function (e.g., 20% loss), and the Gaussian splatting engine may generate a first set of splats that represent a 3D model of the object or scene in the input data at a first fidelity corresponding to the first amount of acceptable loss. 3D streaming system **100** may then configure the loss function with a lower second amount of acceptable loss (e.g., 10% loss), and the Gaussian splatting engine may generate a second set of splats that represent the 3D model of the object or scene in the input data at a second fidelity corresponding to the second amount of acceptable loss. Specifically, the second set of splats may be added to the first set of splats to increase the fidelity (e.g., quality and detail) at which the 3D model is rendered. 3D streaming system **100** may continue decreasing the amount of acceptable loss at which the Gaussian splatting engine generates the 3D model, and may associate the newly generated or added splats to an increasing fidelity value.

(30) In some embodiments, 3D streaming system **100** generates (at **304**) points based on the input data to represent the 3D model at the different fidelities. In some such embodiments, 3D streaming system **100** may use photogrammetry or the 3D scan data to generate points at positions in 3D space and may assign color values to the points based on colors from pixels of a 2D image that align with the generated points. 3D streaming system **100** may select a first set of the generated points (e.g., every 3.sup.rd point) to represent the 3D model at a lowest first level-of-detail, may select a second set of the generated points that added to the first set of the generated points represent the 3D model at a next higher second level-of-detail, and may select a third set of the generated that added to the first and second sets of generated points represent the 3D model at a highest third level-of-detail.

(31) To maintain the association between the 3D primitives and the different fidelities, 3D streaming system **100** tags or otherwise associates (at **306**) the different fidelity values to the different 3D primitives associated with the different fidelities represented by the different fidelity values. For instance, each 3D primitive defined with a fidelity value of 1 may be used to render the 3D content at a lowest first level-of-detail. Each 3D primitive defined with a fidelity value of 2 may be rendered with the 3D primitives defined with the fidelity value of 1 to render the 3D content at a next higher second level-of-detail.

(32) 3D streaming system **100** generates (at **308**) a hierarchical representation for the 3D model at the different fidelities. The hierarchical representation may include an octree, binary tree, K-dimensional tree, or other tree structure for identifying and accessing the 3D primitives that are associated with each fidelity. For instance, the grandparent nodes of the tree may represent the 3D primitives at a lower first fidelity, the parent nodes of the tree combined with the grandparent nodes may represent the 3D primitives at a higher second fidelity, and the leaf nodes of the tree combined

with the nodes at the higher levels may represent the 3D primitives at the highest third fidelity. The nodes at each layer or the 3D primitives associated with each fidelity may be arranged in the tree based on the partitioning of the 3D content space into the volumetric spatial unit. In other words, each branch of the hierarchical representation may correspond to a different volumetric spatial unit, and the 3D primitives linked to that branch may be accessed using the index of the corresponding volumetric spatial unit and the desired fidelity.

(33) In this structure, the 3D primitives associated with each lower level node may be combined or rendered with the 3D primitives associated with nodes at a higher level in the tree to generate the 3D model at the different fidelities. In other words, the 3D primitives associated with the lower levels of the tree add or increase detail to the 3D model representation created by the 3D primitives associated with the higher levels of the tree. Accordingly, if a client requests the 3D model at a lowest first fidelity, 3D streaming system **100** accesses the grandparent layer and streams the 3D primitives associated with the nodes at the grandparent layer. If the client then requests the 3D model at the middle second fidelity and 3D streaming system **100** has already streamed the 3D primitives for the lowest first fidelity, 3D streaming system **100** may access the parent layer and stream the 3D primitives associated with the nodes at the parent layer without having to resend the 3D primitives for the lowest first fidelity.

(34) In some embodiments, 3D streaming system **100** generates discrete 3D models to represent the 3D content at different fidelities. For instance, 3D streaming system **100** may generate a first number of larger-sized meshes to represent the 3D content at a low first fidelity and may generate a second greater number of smaller-sized meshes to represent the 3D content at a high second fidelity. In some such embodiments, there is no benefit in combining the first number of larger-sized meshes with the second greater number of smaller-sized meshes as the second greater number of smaller-sized meshes replace rather than add to the first number of larger-sized meshes.

(35) 3D streaming system **100** generates the manifest to list the different fidelities at which each of the volumetric spatial units for the entire 3D space of the 3D content may be requested. The manifest includes the index assigned to each volumetric spatial unit, the different fidelities for each index at which the 3D primitives in each volumetric spatial unit may be accessed, and an implicit or explicit mapping between the indices and the regions of 3D space spanned by those indices. 3D streaming system **100** provides the manifest to a client that requests the 3D content, and the client, by referencing the manifest, may request and receive different volumetric spatial units of the 3D content at different fidelities so that the quality of the 3D content may be maximized despite changing or limited network and/or rendering resources.

(36) FIG. 4 presents a process **400** for the client-controlled adaptive streaming of 3D content in accordance with some embodiments presented herein. Process **400** is implemented by a client device (e.g., client **101**). The client device may include processing, memory, storage, network, and/or other hardware resources for remotely accessing and rendering 3D content. The client device may correspond to a spatial computing device (e.g., a headset or wearable device), a gaming appliance, and/or other network-enabled computing devices.

(37) Process **400** includes issuing (at **402**) a request to access 3D content, and receiving (at **404**) a manifest in response to the request. The manifest contains indices for different volumetric spatial units that partition the 3D space of the 3D content and different fidelities at which the volumetric spatial units may be rendered. In some embodiments, the manifest specifies an amount of data for each volumetric spatial unit at each fidelity. For instance, 3D streaming system **100** sums the data associated with each 3D primitive in the volumetric spatial unit at each fidelity and includes the sum in association with the index for that volumetric spatial unit and the fidelity. In some embodiments, the manifest includes a mapping between each index and the region of 3D space spanned by the volumetric spatial unit represented by that index.

(38) Process **400** includes determining (at **406**) a set of volumetric spatial units that fall within a current field-of-view. The determination (at **406**) may be based on the mapping between the

manifest indices and the regions of 3D space spanned by the volumetric spatial units represented by those indices. The current field-of-view may initially correspond to a default starting field-of-view (e.g., a default position and orientation for a virtual camera in the 3D space of the 3D content). The default starting field-of-view may change in response to user input or user movements within the 3D space of the 3D content.

(39) Process **400** includes determining (at **408**) available resources for streaming and/or rendering the current field-of-view. The client device may determine (at **408**) the available network resources by testing network performance to determine the bandwidth and/or latency associated with sending and receiving data from 3D streaming system **100** (e.g., issuing (at **402**) the request and receiving (at **404**) the manifest). The client device may determine (at **408**) the available streaming and/or rendering resources based on the fidelity at which the volumetric spatial units for a last field-of-view were received and the buffer between the last field-of-view and the current field-of-view. For instance, if the client device received the last field-of-view at the lowest fidelity with some amount of time remaining before having to render the current field-of-view, the client device may determine that additional resources remain with which to request one or more of the volumetric spatial computing units at a higher fidelity.

(40) Process **400** includes tracking (at **410**) the user focus. In some embodiments, the client device tracks the user focus using eye tracking sensors, head or body movement sensors, controller input, and/or other sensors or user inputs. In some embodiments, the client device tracks the user focus according to the positioning of the set of volumetric spatial units within the current field-of-view or prioritization values associated with the set of volumetric spatial units. For instance, the user focus may be defined or fixed at the center and/or foreground of the current field-of-view. Alternatively, 3D streaming system **100** may set the prioritization values based on 3D primitives of objects-of-interest that are positioned within one or more of the volumetric spatial units and may add the prioritization values to the indices of the one or more volumetric spatial units in the manifest.

(41) Process **400** includes requesting (at **412**) the set of volumetric spatial units in the current field-of-view at different fidelities based on the determined (at **408**) available resources and the determined (at **410**) user focus. The client device may issue a single request that includes the indices for the set of volumetric spatial units and a fidelity value associated each index that corresponds to the fidelity requested for the volumetric spatial unit associated with the index.

(42) If the available resources support receiving the set of volumetric spatial units at the highest fidelity, then the client device requests (at **412**) the set of volumetric spatial units at the highest fidelity. If the available resources cannot support receiving the set of volumetric spatial units at the highest fidelity, then the client device requests (at **412**) the set of volumetric spatial units to preserve the highest fidelity for the volumetric spatial units that receive the user focus and to reduce the fidelity for the volumetric spatial units that are further and further away from the volumetric spatial units that receive the user focus until the total data for requested (at **412**) set of volumetric spatial units is within the determined (at **408**) resources.

(43) In some embodiments, the client device may request the set of volumetric spatial units for the current field-of-view at the lowest fidelity. If there are sufficient available resources remaining with which to render the current field-of-view, the client device may request the 3D primitives at the next higher fidelity for the volumetric spatial units of the current field-of-view that receive the user focus, and may add the 3D primitives at the next higher fidelity to the 3D primitives at the lowest fidelity to progressively increase the fidelity of the 3D content within the time window for the current field-of-view.

(44) Process **400** includes receiving (at **414**) the 3D primitives (e.g., meshes, points, Gaussian splats, etc.) that are defined at the requested fidelity for each volumetric spatial unit of the requested (at **412**) set of volumetric spatial units. Process **400** includes rendering (at **416**) the received (at **414**) 3D primitives to generate a visualization of the 3D content with differing levels-of-detail and/or detail in the current field-of-view.

(45) Process **400** includes detecting (at **418**) a change from the current field-of-view to a next field-of-view. For instance, a user provides input that changes the position or perspective at which the 3D content is viewed. The next field-of-view may include one or more volumetric spatial units that were not in the current field-of-view, may move one or more volumetric spatial units from the current field-of-view out of the next field-of-view, and/or may reposition some volumetric spatial units from the current field-of-view in the next field-of-view (e.g., move from background to foreground, from periphery to the center, etc.).

(46) Process **400** includes determining (at **420**) the new volumetric spatial units in the next field-of-view that were not in the current field-of-view and a change in the user focus from the current field-of-view to the next field-of-view. The determination (at **420**) of the new volumetric spatial units is performed to avoid re-requesting the 3D primitives for the volumetric spatial units that were in the current field-of-view and that remain in the next field-of-view. By determining (at **420**) the new volumetric spatial units and any change in the user focus, the client device may preserve the resources to increase fidelity for the new volumetric spatial units and the volumetric spatial units that receive the user focus in the next field-of-view.

(47) Process **400** includes requesting (at **422**) the new volumetric spatial units in the next field-of-view and/or volumetric spatial units from the current field-of-view that now receive the user focus in the next field-of-view at different fidelities based on the determined (at **408**) available resources and the determined (at **420**) change in the user focus. Once again, the client may specify the index for each requested (at **422**) volumetric spatial unit and the fidelity desired for that volumetric spatial unit.

(48) Process **400** includes receiving (at **424**) the 3D primitives for the new volumetric spatial units and/or additional 3D primitives for increasing fidelity of previously requested volumetric spatial units that receive the user focus in the next field-of-view, and rendering (at **426**) the next field-of-view based on the received (at **424**) 3D primitives and previously received 3D primitives for the requested volumetric spatial units. In this manner, the client is provided control over the adaptive streaming. Consequently, 3D streaming system **100** may provide a stateless streaming service. Also, with the client controlling which volumetric spatial units are requested at which fidelity, latency and/or delay associated with the client reporting changes to the field-of-view to 3D streaming system **100** are eliminated.

(49) FIG. 5 illustrates an example of the client-controlled adaptive streaming of new 3D primitives for a current field-of-view to combine with previously requested 3D primitives from a last field-of-view that remain in the current field-of-view in accordance with some embodiments presented herein. Prior to requesting the last field-of-view, client **101** receives (at **502**) the manifest for the different partitioned volumetric spatial units representing the 3D content and the different fidelities at which the 3D primitives are available in each volumetric spatial unit.

(50) For the last field-of-view, client **101** requests (at **504**) volumetric spatial units with indices 3, 5, and 6 and a lowest first fidelity from 3D streaming system **100**. In response to the request, 3D streaming system **100** traverses (at **506**) the branches of the hierarchical representation for the 3D content that are associated with the identified indices down to a first level associated with the requested lowest first fidelity, and presents (at **508**) the 3D primitives associated with the traversed nodes of the hierarchical representation to client **101**. Client **101** renders (at **510**) the last field-of-view based on the presented (at **508**) 3D primitives.

(51) Client **101** tracks a change from the last field-of-view to the current field-of-view, determines that the current field-of-view removes the volumetric spatial unit identified with index 3, adds volumetric spatial units identified with index 9, and shifts user focus on the volumetric spatial units identified with indices 5 and 6 so that these volumetric spatial units should be presented at a greater fidelity. Client **101** determines that it has already received the lowest fidelity encoding of 3D primitives for the volumetric spatial units identified with indices 5 and 6 and so requests (at **512**) the 3D primitives associated with the second and third fidelities for volumetric spatial units with

indices 5 and 6 and the 3D primitives associated with the lowest first fidelity for volumetric spatial unit identified with index 9.

(52) 3D streaming system **100** traverses (at **514**) the branches of the hierarchical representation based on the fidelities for the identified indices, and presents (at **516**) the 3D primitives associated with the traversed nodes of the hierarchical representation to client **101**. Client **101** combines the newly received 3D primitives with the previously received 3D primitives for volumetric spatial units identified with indices 5 and 6 to render (at **518**) the current field-of-view using the 3D primitives associated with the first, second, and third fidelities for volumetric spatial units identified with indices 5 and 6 and the 3D primitives associated with the first fidelity for volumetric spatial unit identified with index 9.

(53) In some embodiments, client **101** may predict changes to the field-of-view in advance of the changes so as to prefetch the volumetric spatial units at the desired fidelities from 3D streaming system **100**. The predictions may be based on continuing a current movement in the 3D space, detecting interesting or prioritized volumetric spatial units coming into the field-of-view, and/or tracked viewing behaviors of other users.

(54) FIG. **6** illustrates an example of the client-controlled adaptive streaming based on predicted user focus in accordance with some embodiments. Client **101** tracks (at **602**) changes to the user focus to establish the current field-of-view. For instance, client **101** tracks movement of the user gaze via eye tracking sensors, movements of the user head with inertial, rotational, or other sensors, or changes to the user focus based on controller inputs.

(55) Client **101** predicts (at **604**) prioritized volumetric spatial units in a next field-of-view based on the tracked (at **602**) changes to the user focus. The prioritized volumetric spatial units correspond to the volumetric spatial units that are predicted (at **604**) to receive the user focus in the next field-of-view based on the tracked (at **602**) changes to the user focus.

(56) Predicting (at **604**) the next field-of-view may include continuing with the change in the user focus or the associated movement in order to determine the next volumetric spatial units that are likely to receive the user focus. Predicting (at **604**) the next field-of-view may include detecting the features or objects that previously received the user focus and identifying those or similar features or objects in the volumetric spatial units of the current field-of-view or that are just outside the current field-of-view.

(57) Client **101** requests (at **606**) the prioritized volumetric spatial units at a fidelity that is predicted based on the continued change of the user focus and that is supported by the available resources prior to the current field-of-view changing to the next field-of-view or the user focus changing from the current field-of-view. Client **101** receives (at **608**) the 3D primitives for the prioritized volumetric spatial units at the requested fidelities. Client **101** renders (at **610**) the 3D primitives for the prioritized volumetric spatial units immediately in response to the current field-of-view changing to the predicted next field-of-view. In other words, client **101** may render the predicted next field-of-view once the change to the predicted next field-of-view is detected without waiting to submit a request for the 3D primitives in the predicted next field-of-view or receive the 3D primitives.

(58) The client-controlled adaptive streaming may apply to the viewing of static 3D content as well as dynamic or animated 3D content. Dynamic 3D content may include 3D content that changes shape, form, or visual characteristics in response to various events. For instance, 3D characters in a video game may change in response to inputs or interactions with other elements of the video game. Animated 3D content may include 3D content that changes in a predefined manner over time.

(59) To handle dynamic or animated 3D content, the client-controlled adaptive streaming may be modified to include temporal values and/or modified tags. The temporal values may be used to request the changing 3D primitives in a given volumetric spatial unit at a specified fidelity over a period of time. The modified tags may be added to the manifest or sent separate from the manifest

and may be used to notify the client of volumetric spatial units that have been updated or have changed 3D primitives.

(60) FIG. 7 illustrates an example of the client-controlled adaptive streaming with temporal values in accordance with some embodiments presented herein. Client **101** issues (at **702**) a request for 3D content. In this example, the 3D content is dynamic or animated. The dynamic or animated 3D content changes shape, form, and visual characteristics over time based on user input, interactions with other objects in the 3D space, and/or predefined animations for the 3D content.

(61) Client **101** receives (at **704**) the manifest for the 3D content. The manifest specifies the partitioning of the 3D content space into indexed volumetric spatial units and the different fidelities at which the 3D primitives in the indexed volumetric spatial units may be requested.

(62) Client **101** requests (at **706**) the volumetric spatial units that are in a current field-of-view at one or more different fidelities by specifying the indices for the volumetric spatial units in the current field-of-view and the fidelity value associated with each requested volumetric spatial unit in the request. Client **101** also includes temporal values for the requested (at **706**) volumetric spatial units. The temporal values may specify a start time and/or duration for receiving the requested (at **706**) volumetric spatial units at the specified fidelity. In some embodiments, client **101** may specify one set of temporal values for all volumetric spatial units specified in the request. In some other embodiments, client **101** may specify temporal values for each volumetric spatial unit. For instance, client **101** may request a first volumetric spatial unit at a first fidelity for a first duration, may request the first volumetric spatial unit at a second fidelity for a second duration after the first duration, may request a second volumetric spatial unit at the second fidelity for a third duration, and may request the second volumetric spatial unit at the first fidelity for a fourth duration that is after the third duration. 3D streaming system **100** streams (at **708**) the 3D primitives from the requested volumetric spatial units at the fidelities specified for the corresponding temporal values.

(63) Client **101** may set the temporal values based on an analysis of the requested 3D content and/or an amount of change occurring within one or more analyzed frames. For example, the requested 3D content may include a 3D game with lots of motion in the center or foreground and with static 3D elements in the background. In this example, client **101** may request the volumetric spatial units for the center or foreground with a short duration to allow for rapid fidelity changes due to frequent increases and/or decreases in the number of 3D primitives at the center or foreground. Client **101** may request the volumetric spatial units for the background with a long duration because user focus will not be on the background elements and the background elements include static 3D elements. Similarly, for a predefined 3D animation, client **101** may request the volumetric spatial units with temporal values set according to the 3D animation frame rate, the frequency with which the scenes change, and/or the tracking of the user focus. If the user focus changes frequently between 3D primitives in different volumetric spatial units, client **101** may request those volumetric spatial units with a shorter duration. If the user focus remains fixed at the center of the 3D animation, client **101** may request the volumetric spatial units around the center with a longer duration.

(64) 3D streaming system **100** may also notify client **101** when the 3D primitives in a volumetric spatial unit change as these changes may draw the user focus and may be cause for client **101** to request the changing volumetric spatial units at different fidelities. In some embodiments, 3D streaming system **100** may set a flag for each volumetric spatial unit in which the 3D primitives change. The changes may include adding, removing, or repositioning the 3D primitives. The changes may be caused by user inputs, interactions with other objects, and/or predefined animations that affect the 3D primitives in those volumetric spatial units. 3D streaming system **100** may set the flag for a changed volumetric spatial unit in the manifest or may send the flag with the associated volumetric spatial unit index in a separate message to client **101**.

(65) FIG. 8 illustrates an example of the client-controlled dynamic streaming with flagged volumetric spatial units in accordance with some embodiments presented herein. 3D streaming

system **100** generates (at **802**) the volumetric spatial units for a 3D animation, and sends (at **804**) a manifest with the volumetric spatial unit indices and different fidelities at which the 3D primitives of each volumetric spatial unit are encoded to client **101** in response to client **101** requesting the 3D animation.

(66) Client **101** receives the manifest. Client **101** requests (at **806**) the volumetric spatial units for viewing the 3D animation from a particular field-of-view with higher fidelities for the volumetric spatial units receiving the user focus and lower fidelity for the volumetric spatial units that do not receive the user focus. 3D streaming system **100** transmits (at **808**) the 3D primitives in the requested volumetric spatial units at the requested fidelities.

(67) 3D streaming system **100** compares (at **810**) a last frame of the 3D animation to a current frame in order to determine and flag the volumetric spatial units that change between frames. Alternatively, 3D streaming system **100** may generate the 3D animation in response to received user controls or inputs, physics calculation, collisions and/or other interactions between objects in the 3D animation, and/or predefined effects or changes, and may flag the volumetric spatial units that change when generating different frames of the 3D animation. 3D streaming system **100** provides (at **812**) the indices for the flagged or changed volumetric spatial units to client **101**.

(68) Client **101** may determine that the changes may draw the user focus to the changed volumetric spatial units, and may request (at **814**) the changed volumetric spatial units at a different fidelity than in previous requests. In other words, client **101** may request (at **814**) the changed volumetric spatial units at a greater fidelity prior to the user focus shifting to the corresponding regions in the presented 3D animation.

(69) 3D streaming system **100** streams (at **816**) the changed volumetric spatial units at the requested fidelity to client **101**. Client **101** updates the visualization of the 3D animation by rendering the 3D primitives that are encoded in the 3D space of the changed volumetric spatial units at the requested fidelity.

(70) FIG. **9** is a diagram of example components of device **900**. Device **900** may be used to implement one or more of the tools, devices, or systems described above (e.g., 3D streaming system **100**, client **101**, etc.). Device **900** may include bus **910**, processor **920**, memory **930**, input component **940**, output component **950**, and communication interface **960**. In another implementation, device **900** may include additional, fewer, different, or differently arranged components.

(71) Bus **910** may include one or more communication paths that permit communication among the components of device **900**. Processor **920** may include a processor, microprocessor, or processing logic that may interpret and execute instructions. Memory **930** may include any type of dynamic storage device that may store information and instructions for execution by processor **920**, and/or any type of non-volatile storage device that may store information for use by processor **920**.

(72) Input component **940** may include a mechanism that permits an operator to input information to device **900**, such as a keyboard, a keypad, a button, a switch, etc. Output component **950** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more LEDs, etc.

(73) Communication interface **960** may include any transceiver-like mechanism that enables device **900** to communicate with other devices and/or systems. For example, communication interface **960** may include an Ethernet interface, an optical interface, a coaxial interface, or the like.

Communication interface **960** may include a wireless communication device, such as an infrared (IR) receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **900** may include more than one communication interface **960**. For instance, device **900** may include an optical interface and an Ethernet interface.

(74) Device **900** may perform certain operations relating to one or more processes described above. Device **900** may perform these operations in response to processor **920** executing software

instructions stored in a computer-readable medium, such as memory **930**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **930** from another computer-readable medium or from another device. The software instructions stored in memory **930** may cause processor **920** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(75) The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

(76) The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

(77) For example, while series of messages, blocks, and/or signals have been described with regard to some of the above figures, the order of the messages, blocks, and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

(78) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

(79) Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

(80) To the extent the aforementioned embodiments collect, store or employ personal information provided by individuals, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage and use of such information may be subject to consent of the individual to such activity, for example, through well-known “opt-in” or “opt-out” processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

(81) Some implementations described herein may be described in conjunction with thresholds. The term “greater than” (or similar terms), as used herein to describe a relationship of a value to a threshold, may be used interchangeably with the term “greater than or equal to” (or similar terms). Similarly, the term “less than” (or similar terms), as used herein to describe a relationship of a value to a threshold, may be used interchangeably with the term “less than or equal to” (or similar terms). As used herein, “exceeding” a threshold (or similar terms) may be used interchangeably with “being greater than a threshold,” “being greater than or equal to a threshold,” “being less than

a threshold,” “being less than or equal to a threshold,” or other similar terms, depending on the context in which the threshold is used.

(82) No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A method comprising: issuing a request for three-dimensional (3D) content; receiving a manifest in response to the request, wherein the manifest comprises identifiers for different partitioned regions of the 3D content; determining a set of regions from the different partitioned regions that are within a current field-of-view; requesting the identifiers for a first subset of regions from the set of regions at a first fidelity and the identifiers for a second subset of regions from the set of regions at a second fidelity; and generating the current field-of-view with the first subset of regions rendered at the first fidelity and the second subset of regions rendered at the second fidelity.
2. The method of claim 1 further comprising: determining that the first subset of regions correspond to background or static regions of the 3D content and that the second subset of regions correspond to foreground or changing regions of the 3D content; and wherein requesting the identifiers comprises: requesting the identifiers for the first subset of regions at the first fidelity with a first temporal value; and requesting the identifiers for the second subset of regions at the second fidelity with a second temporal value that specifies a shorter time or duration than the first temporal value.
3. The method of claim 2 further comprising: receiving primitives representing the background or static regions of the 3D content at the first fidelity for a first duration defined by the first temporal value; and receiving primitives representing the foreground or changing regions of the 3D content at the second fidelity for a second duration defined by the second temporal value, wherein the second fidelity corresponds to a higher resolution or level-of-detail than the first fidelity.
4. The method of claim 2 further comprising: requesting the identifiers for the second subset of regions at a third fidelity that is different than the second fidelity with a third temporal value; receiving primitives of the 3D content defined in the foreground or changing regions at the third fidelity for a third duration defined by the third temporal value; and generating a field-of-view during the third duration with the first subset of regions rendered at the first fidelity and the second subset of regions rendered at the third fidelity.
5. The method of claim 1 further comprising: detecting an amount of change occurring in the first subset of regions; and wherein requesting the identifiers comprises: requesting the identifiers for the first subset of regions at the first fidelity with a first temporal value and at a third fidelity with a second temporal value in response to detecting change in the first subset of regions.
6. The method of claim 1 further comprising: tracking user focus on the second subset of regions in the current field-of-view; requesting the first subset of regions at the first fidelity for fields-of-view over a first duration in response to the user focus being on the second subset of regions; and requesting the second subset of regions at the second fidelity for field-of-views over a second duration that is less than the first duration in response to the user focus being on the second subset of regions.
7. The method of claim 1 further comprising: tracking user focus changing from the second subset

of regions to the first subset of regions from the current field-of-view to a next field-of-view; requesting the identifiers for the first subset of regions at the second fidelity and the identifiers for the second subset of regions at the first fidelity, wherein the second fidelity is greater than the first fidelity; and generating the next field-of-view with the first subset of regions rendered at the second fidelity and the second subset of regions rendered at the first fidelity.

8. The method of claim 1 further comprising: detecting changes to the 3D content that occur in the first subset of regions and not in the second subset of regions; and requesting the identifiers for the first subset of regions at different fidelities with different temporal values that span a particular duration and the identifiers for the second subset of regions at the second fidelity with a single temporal value that spans the particular duration.

9. The method of claim 1 further comprising: receiving an update to the manifest that indicates a change in a particular region of the first subset of regions, wherein the particular region is associated with a particular identifier; and requesting the particular identifier at a third fidelity in response to receiving the update.

10. The method of claim 1 further comprising: analyzing changes occurring throughout different frames of the 3D content; determining that the changes affect the first subset of regions; and wherein requesting the identifiers comprises: requesting the identifiers for the first subset of regions at different fidelities with different temporal values that collectively span a particular duration in response to determining that the changes affect the first subset of regions; and requesting the identifiers for the second subset of regions at the second fidelity with a second temporal value that spans the particular duration in response to determining that the changes affect the first subset of regions.

11. The method of claim 1, wherein the manifest further comprises values corresponding to different fidelities at which the 3D content is encoded, and wherein requesting the identifiers comprises: requesting the identifiers for the first subset of regions with a first value from the values of the manifest that corresponds to the first fidelity; and requesting the identifiers for the second subset of regions with a second value from the values of the manifest that corresponds to the second fidelity.

12. The method of claim 1 further comprising: receiving a first set of primitives that encode parts of the 3D content in the first subset of regions at the first fidelity in response to said requesting; and receiving a second set of primitives that encode parts of the 3D content in the second subset of regions at the second fidelity in response to said requesting.

13. The method of claim 12, wherein the first set of primitives and the second set of primitives include one or more meshes, points, or splats that are defined in a 3D space of the 3D content.

14. The method of claim 1 further comprising: mapping the identifiers of the manifest to positions of the different partitioned regions in a 3D space; and selecting the identifiers of the first subset of regions and the identifiers of the second subset of regions that map to the positions of the first subset of regions and the second subset of regions within the current field-of-view.

15. A device comprising: one or more hardware processors configured to: issue a request for three-dimensional (3D) content; receive a manifest in response to the request, wherein the manifest comprises identifiers for different partitioned regions of the 3D content; determine a set of regions from the different partitioned regions that are within a current field-of-view; request the identifiers for a first subset of regions from the set of regions at a first fidelity and the identifiers for a second subset of regions from the set of regions at a second fidelity; and generate the current field-of-view with the first subset of regions rendered at the first fidelity and the second subset of regions rendered at the second fidelity.

16. The device of claim 15, wherein the one or more hardware processors are further configured to: determine that the first subset of regions correspond to background or static regions of the 3D content and that the second subset of regions correspond to foreground or changing regions of the 3D content; and wherein requesting the identifiers comprises: requesting the identifiers for the first

subset of regions at the first fidelity with a first temporal value; and requesting the identifiers for the second subset of regions at the second fidelity with a second temporal value that specifies a shorter time or duration than the first temporal value.

17. The device of claim 16, wherein the one or more hardware processors are further configured to: receive primitives representing the background or static regions of the 3D content at the first fidelity for a first duration defined by the first temporal value; and receive primitives representing the foreground or changing regions of the 3D content at the second fidelity for a second duration defined by the second temporal value, wherein the second fidelity corresponds to a higher resolution or level-of-detail than the first fidelity.

18. The device of claim 16, wherein the one or more hardware processors are further configured to: request the identifiers for the second subset of regions at a third fidelity that is different than the second fidelity with a third temporal value; receive primitives of the 3D content defined in the foreground or changing regions at the third fidelity for a third duration defined by the third temporal value; and generate a field-of-view during the third duration with the first subset of regions rendered at the first fidelity and the second subset of regions rendered at the third fidelity.

19. The device of claim 15, wherein the one or more hardware processors are further configured to: detect an amount of change occurring in the first subset of regions; and wherein requesting the identifiers comprises: requesting the identifiers for the first subset of regions at the first fidelity with a first temporal value and at a third fidelity with a second temporal value in response to detecting change in the first subset of regions.

20. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of a client device, cause the client device to perform operations comprising: issuing a request for three-dimensional (3D) content; receiving a manifest in response to the request, wherein the manifest comprises identifiers for different partitioned regions of the 3D content; determining a set of regions from the different partitioned regions that are within a current field-of-view; requesting the identifiers for a first subset of regions from the set of regions at a first fidelity and the identifiers for a second subset of regions from the set of regions at a second fidelity; and generating the current field-of-view with the first subset of regions rendered at the first fidelity and the second subset of regions rendered at the second fidelity.
